SAMPLE

The EPR Paradox

Einstein's Challenge to Quantum Mechanics

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Introduction

In the spring of 1935, a short paper appeared in an American physics journal carrying an argument that would trouble physics for the rest of the century. Its three authors, Albert Einstein, Boris Podolsky, and Nathan Rosen, asked a question in their title: can the quantum-mechanical description of physical reality be considered complete? Their answer, argued with sharpened precision, was no. Quantum mechanics, they claimed, either fails to describe certain real properties of the world or else attributes to nature a connectedness so strange that measuring one particle appears to fix the state of another far away, instantly, without anything passing between them. The

paper became famous as the EPR paradox, and for nearly three decades it was treated as a curiosity of philosophy, a puzzle for the seminar room with no experimental weight behind it.

That judgment turned out to be spectacularly wrong. In 1964, a young theorist named John Bell, working on quantum foundations as a private hobby while his day job concerned the design of particle accelerators, discovered something the founders had missed: the philosophical dispute could be converted into a numerical inequality, a testable prediction about the strength of correlations, on which quantum mechanics and any theory of the Einstein type would disagree. The universe, Bell showed, had to choose. By the 1970s experimenters had begun to ask it to choose, and by 2015 the last of the experimental loopholes had been closed. The verdict, now as firm as anything in science, went against Einstein's side. Nature really does entertain correlations that cannot be explained by any local theory of pre-existing properties.

And yet the story is not a simple defeat of the greatest physicist of the age. The EPR paper forced physics to articulate, for the first time, what it means to call a theory complete, what it means for something to be real, and what it means for two events to influence one another. The distinction Bell later drew between questions of interpretation and questions of prediction came directly out of that clearing of ground. The very correlations EPR treated as a reductio ad absurdum became, in the hands of later generations, a resource: the basis of quantum cryptography that can certify its own security, of quantum teleportation, of entanglement- based networks now being built across continents and from satellites. The paradox that began as an attack on quantum mechanics ended by handing physics and engineering one of its most valuable tools. Few documents in the history of science have been so wrong in their aim and so right in their effect.

This book follows the whole arc. It begins with the debates between Einstein

and Niels Bohr that made the issue unavoidable, walks through the argument of the 1935 paper itself in plain language, and follows the responses: Bohr's famous reply, Schrödinger's coinage of entanglement, Bell's theorem and the decades-long experimental campaign it ignited. It explains what the experiments actually refute, which is subtler than the usual shorthand suggests, and what they leave untouched, including the iron rule that no usable signal travels faster than light. The closing chapters take up the modern landscape: the interpretations still contending to make sense of the results, the technologies built from the very correlations EPR found absurd, and the question their argument raised that remains as open today as it was in 1935.

No equations appear in these pages. The argument of EPR, Bell's inequality, and the logic of the loophole-free experiments can all be told honestly in words, and they deserve to be told that way, because the subject is, at bottom, not a mathematical curiosity but a question about the character of the world: whether the properties of things exist before we look, whether influence requires contact, and what it cost physics to learn that the answers are not the ones common sense would have guessed.

Chapter 1 begins in the smoke of the Solvay conferences, with two friends arguing about the same equation and disagreeing about what it meant.

Chapter 1: The Discovery of the EPR Paradox

The paper of 1935 did not appear out of nowhere. It was the last and sharpest move in an argument that Einstein had been conducting with the quantum theory for nearly a decade, and to understand what its authors were trying to do, the argument has to be reconstructed.

Einstein had been one of quantum theory's midwives. His 1905 paper on the photoelectric effect proposed that light comes in lumps, a suggestion even more radical than Max Planck's original quantization and one that helped earn Einstein the Nobel Prize. He did not object to quantum mechanics because he found it strange or because he could not follow it. His objection was more specific and more interesting: he believed the theory was incomplete. By incomplete he meant something technical. A complete theory, as he framed it in later statements of the position, is one in which every element of physical reality has a counterpart in the theory. If an experiment can be performed whose outcome is certain without the system being disturbed, then there exists something real about the system corresponding to that certainty, and a

theory that does not represent it is missing part of the world.

Bohr answered Einstein's challenges at the famous Solvay meetings of 1927 and 1930, and by most accounts he won those exchanges. The photon-in-a-box thought experiment of 1930, in which Einstein proposed weighing a box of light to determine the energy and timing of an emitted photon with arbitrary precision, was dismantled by Bohr using general relativity itself, an argument so elegant that Einstein conceded the point. But conceding individual rounds did not change his underlying conviction, and through the late 1920s and early 1930s his letters show him searching for a version of the puzzle that could not be dissolved by Bohr's artistry, because it would be a fact about the theory rather than a fact about cleverly arranged measuring devices.

Two things then happened in quick succession. Einstein emigrated, taking a

position at the Institute for Advanced Study in Princeton, and there he acquired two younger collaborators, Boris Podolsky, a Russian-born theorist, and Nathan Rosen, a young American postdoctoral researcher. Rosen had already made his name in general relativity, where the object now called a wormhole was first examined in a paper by Einstein and Rosen. The three set about turning Einstein's conviction into a published argument, and the construction they produced, submitted in the spring of 1935, ran roughly as follows in plain language.

Consider a pair of particles that have interacted and then separated, one sent to a laboratory here, the other to a laboratory far away. The laws of quantum mechanics, applied to the pair, predict perfect correlations between certain measurements on the two: measure one particle's position and you can predict the distant particle's position with certainty; measure the momentum of the first and you can predict the distant momentum with certainty. Neither measurement touches the distant particle. Now apply the criterion of reality: if the distant particle's position can be predicted with certainty without disturbing it, its position is an element of reality; the same reasoning makes its momentum an element of reality. Both properties are therefore real at once. But quantum mechanics forbids assigning sharp values to both position and momentum simultaneously; that is the content of the uncertainty principle. Hence quantum mechanics fails to represent real properties that the world possesses, and is therefore incomplete.

Podolsky, by most accounts, polished the writing, and the published introduction frames the argument with a generality Einstein thought excessive. Einstein wrote afterward to his friend Erwin Schrödinger, who was then in Oxford, that the main point had been smothered by the paper's erudition. The essential question, he said, was not the elaborate packaging but a simple one:

can the state of one system be considered complete when its description refers to correlations with a distant system? Schrödinger answered that letter

with a burst of creativity, giving the strange correlation its enduring name, entanglement, in his reply and the papers that followed it that summer. The same months produced his famous cat, proposed in part as a second, more vivid demonstration of the same uncomfortable territory: quantum description that fails to be a description of a definite world.

The paper appeared in May 1935 under the title its question became known by, in the Physical Review, and it drew an immediate reply from Bohr, published the same year in the same journal under the same title, that historians still find easier to admire than to summarize. What matters for the story is what the physics community did with the exchange, which was very little, and for understandable reasons. Quantum mechanics was conquering

one field after another; nuclear physics was exploding, literally, in the same decade; and a dispute about the meaning of the word reality, with no experiment proposed to settle it, was easy to shelve. Einstein returned to unified field theory. Podolsky went to industry and academia. Rosen had a long career in physics. The paradox entered the textbooks as a footnote, a famous objection, duly noted and left unresolved.

The footnote waited twenty-nine years for its revenge, and the revenge, when it came, did not come from the philosophical side of the argument. It came from Bell, and it converted the whole dispute into an experiment. Before turning to that conversion, it is worth being precise about what the paradox actually claims, because the popular version, involving faster- than-light signals and particles communicating across space, is wrong in ways that matter. That precision is the business of the next chapter.

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